**FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 4**

**B.Sc. MATHEMATICS HONOURS**

**MAT5CJ303: COMPLEX ANALYSIS I**

Modules III–IV: Cauchy–Riemann Equations (Polar Form), Harmonic Functions, Exponential, Logarithmic, Trigonometric and Hyperbolic Functions

| **Maximum Time: 2 Hours** | | **Maximum Marks: 70** |
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| | **Section A** | |
| | **All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)** | |
| 1 | State the sufficient conditions for differentiability of $f(z) = u(x,y) + iv(x,y)$ at a point $z$. Use them to show that $f(z) = x^2 + y^2 + 2ixy$ is differentiable on the $x$-axis but is nowhere analytic. | **3** |
| 2 | Write the Cauchy–Riemann equations in polar coordinates. Use them to show that $f(z) = 5r\cos\theta + r^4\cos 4\theta + i(5r\sin\theta + r^4\sin 4\theta)$ is analytic in the domain $r > 0$, and find $f'(z)$. | **3** |
| 3 | Define a harmonic function in a domain $D$. Verify that $u(x,y) = x^3 - 3xy^2$ is harmonic in the entire complex plane. | **3** |
| 4 | Define a harmonic conjugate of a harmonic function $u(x,y)$. Show that $v(x,y) = 2xy$ is a harmonic conjugate of $u(x,y) = x^2 - y^2$, and write the analytic function $u + iv$ in terms of $z$. | **3** |
| 5 | Define the complex exponential function $e^z$. Show that $e^z$ is periodic with the pure imaginary period $2\pi i$, and explain why the real function $e^x$ has no such period. | **3** |
| 6 | For $z = x + iy$, find $\lvert e^{z^2}\rvert$ and all values of $\arg(e^{z^2})$ in terms of $x$ and $y$. Also show that $\overline{e^z} = e^{\bar{z}}$ | **3** |
| 7 | Define the complex logarithm $\ln z$ and its principal value $\operatorname{Ln} z$. Find all values of $\ln(-2+2i)$ and the principal value $\operatorname{Ln}(-2+2i)$. | **3** |
| 8 | Find all complex solutions of the equation $e^z = 4i$. Which of them is $\operatorname{Ln}(4i)$? | **3** |
| 9 | Define the complex sine and cosine functions. Using $\cos z = \cos x\cosh y - i\sin x\sinh y$, express $\cos(2-4i)$ in the form $a + ib$. | **3** |
| 10 | Define the complex hyperbolic sine and cosine functions. Show that $\sinh z$ and $\cosh z$ are entire, and find their derivatives. | **3** |
| | **Section B** | |
| | **All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)** | |
| 11 | State the criterion for analyticity (Theorem 3.3.2). Use it to show that $f(z) = \dfrac{x}{x^2+y^2} - i\dfrac{y}{x^2+y^2}$ is analytic in any domain $D$ that does not contain $z = 0$, and find $f'(z)$ using $f'(z) = \dfrac{\partial u}{\partial x} + i\dfrac{\partial v}{\partial x}$. | **6** |
| 12 | Let $x = r\cos\theta$, $y = r\sin\theta$ and $f(z) = u(x,y) + iv(x,y) = u(r,\theta) + iv(r,\theta)$. Starting from the chain rule, derive the Cauchy–Riemann equations in polar form, $\dfrac{\partial u}{\partial r} = \dfrac{1}{r}\dfrac{\partial v}{\partial \theta}$, $\dfrac{\partial v}{\partial r} = -\dfrac{1}{r}\dfrac{\partial u}{\partial \theta}$, and show that $f'(z) = e^{-i\theta}\left(\dfrac{\partial u}{\partial r} + i\dfrac{\partial v}{\partial r}\right)$. | **6** |
| 13 | Verify that $u(x,y) = e^{-x}(x\sin y - y\cos y)$ is harmonic in the entire complex plane. Find a harmonic conjugate $v$ of $u$, and express the analytic function $f = u + iv$ in terms of $z$. | **6** |
| 14 | Prove that if $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain $D$, then $u$ and $v$ are harmonic in $D$ (assume the second-order partial derivatives of $u$ and $v$ are continuous in $D$). | **6** |
| 15 | Using the Cauchy–Riemann equations, show that $e^z$ is entire and that $\dfrac{d}{dz}e^z = e^z$. Also prove that $e^{z_1}e^{z_2} = e^{z_1+z_2}$ and $\dfrac{e^{z_1}}{e^{z_2}} = e^{z_1-z_2}$. | **6** |
| 16 | Prove that $\ln(z_1z_2) = \ln z_1 + \ln z_2$ for nonzero complex numbers $z_1, z_2$, interpreting both sides as sets of values. Find two nonzero complex numbers $z_1$ and $z_2$ for which $\operatorname{Ln}(z_1z_2) \neq \operatorname{Ln} z_1 + \operatorname{Ln} z_2$. | **6** |
| 17 | Show that $\sin z = \sin x\cosh y + i\cos x\sinh y$ and deduce $\lvert\sin z\rvert^2 = \sin^2 x + \sinh^2 y$. Hence show that the only zeros of $\sin z$ are the real numbers $z = n\pi$, $n = 0, \pm 1, \pm 2, \ldots$ | **6** |
| 18 | Prove that $\sin z = -i\sinh(iz)$ and $\cos z = \cosh(iz)$. Use the definitions to verify $\cosh^2 z - \sinh^2 z = 1$, and find all zeros of $\cosh z$. | **6** |
| | **Section C** | |

| Answer any ONE. Each Question carries 10 marks (1x10=10 Marks) | | |
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| 19 | (a) Find all solutions of the equation $\sin z = 5$, Explain why the equation has no real solution but has infinitely many complex solutions.<br>(b) Find all solutions of the equation $\cosh z = i$. Find $\frac{d}{dz}\tanh z$ and state where $\tanh z$ is analytic. | **10** |
| 20 | (a) Suppose $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain $D$. Prove that (i) if $f'(z) = 0$ in $D$, then $f(z) = c$ in $D$, where $c$ is a complex constant, and (ii) if $\lvert f(z)\rvert$ is constant in $D$, then $f(z)$ is constant in $D$.<br>(b) Solve $e^{2z} + e^z + 1 = 0$, and give the two solutions obtained from the principal value of the logarithm. | **10** |